

NIS, Serbia

WMO: 133880

Lat: 43.337N

Lon: 21.854E

Elev: 648

StdP: 14.36

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	13.8	18.1	8.2	8.6	16.4	12.4	10.6	19.8	25.4	32.5	20.8	32.1	5.0	320	0.457

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	24.4	95.4	69.4	92.4	68.8	89.2	68.3	72.3	88.7	70.8	86.8	69.5	84.7	5.0	340

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
67.4	103.0	78.0	65.9	97.9	76.5	64.4	92.8	75.1	36.2	88.7	35.0	87.3	33.9	85.1	78.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.3	17.1	14.5	7.0	100.8	6.2	4.2	2.6	103.8	-1.1	106.2	-4.5	108.5	-9.1	111.6
			6.3	75.1	6.0	1.6	2.0	76.2	-1.5	77.2	-4.8	78.1	-9.2	79.3
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	54.6	33.9	38.0	45.5	54.6	63.1	69.8	73.6	73.6	64.8	55.2	46.1	36.2
	DBStd	15.80	8.58	9.36	8.32	7.31	6.40	6.09	5.75	5.70	6.81	7.65	8.26	8.24
	HDD50	1712	499	345	188	38	1	0	0	0	0	39	169	433
	HDD65	4648	963	757	605	318	114	23	5	7	86	310	567	893
	CDD50	3401	2	8	48	176	406	595	732	731	445	201	52	5
	CDD65	862	0	0	0	6	54	169	272	272	80	8	1	0
	CDH74	9628	0	0	6	112	615	1830	3025	3058	855	124	3	0
	CDH80	4167	0	0	0	22	150	735	1453	1488	298	21	0	0
Wind	WSAvg	4.9	4.9	5.5	6.0	5.4	4.8	4.6	4.8	4.7	4.5	4.6	4.6	5.0
Precipitation	PrecAvg	24.2	1.7	1.7	1.7	2.0	2.8	2.9	1.8	1.6	1.9	1.6	2.2	2.2
	PrecMax	30.5	3.4	3.7	3.8	4.1	5.4	8.1	3.7	4.6	7.0	3.3	4.2	4.4
	PrecMin	18.1	0.1	0.3	0.4	0.9	0.8	0.8	0.5	0.2	0.3	0.0	0.7	0.0
	PrecStd	3.3	0.9	0.9	1.1	0.9	1.5	1.5	1.0	1.1	1.7	1.1	1.1	1.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	59.6	67.7	74.2	83.0	88.2	95.0	100.8	100.7	93.6	82.5	72.1	62.4
		MCWB	49.8	52.7	55.5	61.4	65.2	70.8	69.1	68.9	67.8	63.3	58.4	52.8
	2%	DB	54.5	60.9	69.5	76.9	84.1	91.3	96.2	96.0	87.7	77.2	66.9	56.9
		MCWB	47.2	49.5	53.0	58.6	64.7	70.1	69.5	69.3	65.4	61.4	55.8	49.3
	5%	DB	50.1	56.8	64.7	72.3	80.8	87.8	91.8	92.5	82.7	73.1	62.9	52.1
		MCWB	44.7	47.5	51.2	56.1	63.9	68.8	69.0	68.3	64.3	60.2	54.0	46.7
	10%	DB	46.2	51.9	60.6	68.3	77.1	84.2	88.1	88.4	78.7	68.6	59.0	48.1
		MCWB	42.4	45.3	48.9	54.9	62.3	67.8	68.2	67.6	63.0	58.1	51.8	44.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	50.4	53.2	56.5	63.0	68.3	74.4	73.7	73.8	69.2	65.3	59.4	53.5
		MCDB	58.0	64.9	71.3	79.0	82.0	87.9	89.1	90.8	87.1	79.1	69.2	60.0
	2%	WB	47.7	50.3	54.1	60.0	66.3	72.0	72.0	71.7	67.2	62.9	57.0	49.9
		MCDB	53.8	59.5	66.6	73.8	80.0	87.1	88.7	89.0	83.0	74.1	65.8	55.6
	5%	WB	45.0	48.2	51.8	57.8	64.7	70.2	70.6	70.2	65.7	60.9	54.8	47.1
		MCDB	49.5	55.6	62.9	70.2	78.1	84.5	86.9	86.4	79.6	70.7	61.8	51.9
	10%	WB	42.3	45.5	49.7	55.7	63.1	68.6	69.3	68.8	64.1	58.9	52.4	44.3
		MCDB	45.6	51.3	59.7	66.0	75.3	82.3	84.6	84.2	76.2	67.7	58.5	47.3
Mean Daily Temperature Range	5% DB	MDBR	13.2	16.0	19.4	21.2	21.8	23.0	24.4	25.3	22.4	20.6	17.0	12.3
		MCDBR	20.5	25.3	28.1	29.4	28.3	28.9	31.4	32.7	30.4	28.2	23.7	19.6
		MCWBR	13.8	15.2	15.1	14.4	12.7	11.4	10.8	11.1	12.4	14.3	13.8	13.2
	5% WB	MCDBR	19.6	22.9	25.6	25.3	25.3	26.2	26.9	27.2	26.2	25.2	21.7	18.3
		MCWBR	13.5	14.0	14.3	12.9	12.2	11.4	10.3	10.6	11.3	13.3	12.8	12.6
Clear-Sky Solar Irradiance	taub	0.336	0.363	0.403	0.461	0.450	0.440	0.442	0.456	0.433	0.392	0.359	0.334	
	taud	2.333	2.250	2.199	2.056	2.148	2.220	2.213	2.174	2.209	2.314	2.362	2.362	
	Ebn at Noon	247	259	263	256	263	265	263	254	250	246	237	236	
	Edh at Noon	26	34	40	50	47	44	44	44	40	32	26	24	
All-Sky Solar Radiation	RadAvg	449	715	1071	1414	1715	1954	2011	1810	1282	876	543	388	
	RadStd	69	100	140	173	168	178	148	163	114	124	64	58	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
		+1.06	N/A	N/A	N/A	+0.83	+1.08	N/A	N/A	N/A	N/A
Regional (44 neighbors)		+1.10	N/A	N/A	N/A	+0.68	N/A	N/A	-286	N/A	N/A

Nomenclature: See separate page